

AI Adoption in Construction and Facilities: A Qualitative Study of AEC Industry Practices

Jacob Appling

CMET 6000: Spc Topic in Constr & Fac Mgmt

University of North Carolina at Charlotte

May 6, 2026

Abstract

Artificial intelligence is becoming an increasingly important part of construction and facilities management, but adoption across the architecture, engineering, and construction (AEC) industry remains uneven. This study investigates how organizations are currently using, evaluating, and managing AI through qualitative interviews with professionals from general contractors and a healthcare facilities organization. The research connects interview findings with academic literature on AI adoption, the Technology Acceptance Model, organizational readiness, and capability maturity. Findings suggest that AI adoption is most successful when it is problem-driven, leadership-supported, and introduced gradually through pilot programs or small implementation teams. Common use cases included Microsoft Copilot for meeting notes and document review, AI-supported project management platforms, reality capture and safety monitoring, healthcare documentation tools, and internal AI agents for operational support. Across interviews, major barriers included data security, employee trust, training needs, cost, and limited field-side adoption. The results indicate that AI is currently delivering the most value in office-based, document-heavy, and process-driven workflows, while field implementation remains more complex. The study concludes that successful AI adoption in construction and facilities requires strong governance, clear use cases, hands-on training, and measurable performance indicators before broad organizational rollout.

Keywords: artificial intelligence, construction technology, facilities management, AI adoption, qualitative research

Introduction and Literature Review

Artificial intelligence is increasingly influencing how organizations manage information, evaluate risk, coordinate work, and support decision-making. In the architecture, engineering, and construction (AEC) sector, AI adoption is especially important because project work is document-heavy, schedule-driven, safety-sensitive, and dependent on coordination among owners, designers, contractors, subcontractors, and facility operators. Although AI is often discussed as a technical advancement, its successful adoption depends on more than software capability. Organizations must also address leadership support, training, data security, user trust, workflow integration, and measurable value.

The purpose of this report is to examine how AI is being adopted in construction and facilities-related organizations and to compare those practices with academic frameworks on AI adoption and organizational innovation. The study draws on interviews with professionals representing a healthcare facilities organization and two general contractors. The analysis focuses on how these organizations are using AI, what barriers they face, what benefits they are beginning to observe, and what strategies appear most important for responsible implementation.

Current literature frames AI as a major force in the digital transformation of construction and facilities-related work. Adebayo et al. (2025) found that AI applications in construction project management are especially concentrated in planning, monitoring, and control, including forecasting, scheduling, safety monitoring, and decision-support tasks. This suggests that AI adoption in construction is often driven first by practical use cases that can improve project performance or reduce repetitive administrative work. In this study, interview participants similarly described AI as most useful when it addressed specific workflow problems rather than when it was adopted as a general-purpose trend.

Another important framework for understanding AI adoption is the Technology Acceptance Model (TAM). Na et al. (2023) argued that adoption of AI-based construction technologies is shaped by perceived usefulness, perceived ease of use, and compatibility with existing work practices. This is especially relevant to construction because the industry is fragmented, risk-averse, and highly dependent on field experience. A tool may have strong technical capability but still fail if users do not trust it, understand it, or see how it improves their daily work. The interviews in this study strongly support this point, as participants repeatedly emphasized that AI must be useful, integrated into existing platforms, and introduced in a way that does not disrupt daily operations.

The literature also points to organizational readiness and staged maturity as important adoption factors. Omotayo et al. (2025) examined AI-powered BIM adoption through capability maturity modeling and identified dimensions such as adoption and integration, technological advancement, policy support, training, incentives, industry readiness, and cultural attitudes. This framework is useful because it treats adoption as a gradual process rather than a one-time software purchase. The organizations interviewed for this report used similar staged approaches, including governance committees, pilot projects, implementation teams, training sessions, and key performance indicators.

Research on generative AI in construction risk management further highlights the balance between opportunity and risk. Mohamed et al. (2025) found that recent studies identify benefits in technical, operational, technological, and integration areas, while also emphasizing risks related to data, security, performance, and social impacts. These concerns appeared throughout the interviews. Both general contractors and the healthcare organization were interested in AI-

enabled efficiency, but they were also cautious about proprietary data, patient information, compliance, and the reliability of AI-generated outputs.

Taken together, the literature suggests that AI adoption in construction and facilities should be viewed as an organizational innovation process. Purchasing an AI tool is not enough. Adoption depends on leadership, governance, trust, training, workflow fit, and the ability to measure value over time. This report uses those ideas as the foundation for analyzing how interviewed organizations are approaching AI in practice.

Methodology

This study used a qualitative interview approach supported by secondary research from peer-reviewed literature. Semi-structured interviews were conducted with professionals connected to construction, construction technology, healthcare IT, and facilities-related operations. The interview questions followed the assignment questionnaire and addressed role background, current AI use, adoption and implementation, challenges and barriers, observed outcomes, and strategic vision.

The sample included four professional perspectives across three organizations. Participants included an IT director from a healthcare organization, a director of construction technology from a general contractor, a project manager from the same general contractor, and a senior vice president/technology group leader from a second general contractor. Because explicit consent to use employee names and company names was not confirmed for all interviews, the findings are reported using anonymous organizational labels: Healthcare Organization, General Contractor A, and General Contractor B. Individual names are not used.

Data were analyzed using thematic analysis. Interview transcripts and written responses were reviewed to identify recurring patterns across organizations. The findings were then

grouped into themes related to governance, problem-driven adoption, workflow integration, training and cultural resistance, security and data management, office versus field adoption, and measurable value. These themes were compared with the academic literature on TAM, organizational readiness, capability maturity, and AI-related risk management.

This study has limitations. The sample size was small and focused on organizations connected to healthcare and commercial construction in one regional context. Interview responses reflect professional perceptions and selected examples rather than a comprehensive measurement of AI performance. However, the interviews provide useful insight into how AI adoption is currently being evaluated and implemented in real-world AEC and facilities environments.

Organizational Context

Healthcare Organization

The healthcare organization represents an owner/facilities-side perspective. The interviewee was an IT director responsible for applications integration, which includes connecting and supporting systems used across the organization. The organization operates in a highly regulated healthcare environment where patient data, privacy, and compliance are major considerations. Its AI initiatives include vendor-based healthcare tools, ambient clinical documentation, and internally developed AI agents.

General Contractor A

General Contractor A represents a commercial construction firm using technology across project management and field operations. This contractor was kind enough to offer two employees to interview. Interviewees included a director of construction technology and a project manager. The technology role included tools such as drones, 360-degree photography,

laser scanning, BIM coordination, renderings, and AI-related pilot programs. The project management role involved document control, meeting minutes, RFIs, submittals, drawings, scheduling, subcontractor coordination, and support for the field superintendent.

General Contractor B

General Contractor B represents a mid-sized general contracting firm with approximately 350 employees. The interviewee was a senior vice president involved in the organization's technology group and process improvement efforts. The firm is using AI for research, quick answers, subcontract review, internal process support, and early agent-based workflows. Its approach is shaped by leadership support, data security concerns, and gradual implementation.

Findings and Thematic Discussion

Table 1 summarizes the interview findings by organization, role, tools, purpose, challenges, and observed outcomes. The table shows that AI adoption is already occurring across multiple functions, but implementation remains cautious and staged.

Table 1
Comparative Summary of Interview Findings

Organization	Role	AI Tools / Platforms	Purpose / Use Cases	Challenges	Outcomes / Value
Healthcare Organization	IT Director for Applications Integration	Epic AI tools; ambient clinical documentation; Microsoft Copilot Studio; custom internal agents	Provider message support; clinical documentation; IT help desk automation; on-call lookup; workflow support	HIPAA compliance; PHI protection; governance; data security; ensuring strategic fit	Improved provider efficiency; estimated 9 hours/week saved from a computer reboot agent and 9 hours/week from an on-call lookup feature
General Contractor A	Director of Construction Technology	Microsoft Copilot; Read AI; DroneDeploy; 360 photos; drones; laser scans; BIM/reality capture tools	Email and meeting summaries; safety observations; reality capture; AI pilot projects; future progress tracking	Training needs; time constraints; cost; cultural resistance; avoiding flashy tools without practical value	Early pilot learning; improved meeting and documentation support; automated safety observations; potential for progress tracking
General Contractor A	Project Manager	Microsoft Copilot; Procore AI/search; 360 cameras; safety AI tools	Meeting minutes; specification and drawing search; submittal and RFI support; safety monitoring	Learning new software; field adoption; cost; need for training and support	More efficient meetings; faster access to specifications and details; support for identifying safety issues
General Contractor B	Senior VP / Technology Group Leader	Microsoft Copilot; Claude Code; internal AI agents	Research; quick answers; subcontract review;	Data security; proprietary information	Time savings on document-heavy work; reduced time

Organization	Role	AI Tools / Platforms	Purpose / Use Cases	Challenges	Outcomes / Value
			software/process improvement; internal process Q&A	protection; trust in AI output; limited field-side adoption	searching for process answers; early agent-based workflow development

Theme 1: AI Adoption Is Problem-Driven

A major theme across the interviews was that AI adoption should begin with real organizational problems rather than with the technology itself. The construction technology director at General Contractor A stated that AI is not a solution by itself and should be connected to specific pain points, tedious tasks, or automation opportunities. This view was reinforced by General Contractor B, where AI agents are being developed for practical uses such as subcontract review and internal process questions. In the healthcare organization, internal agents were developed around specific IT workflow problems such as rebooting computers and identifying the correct on-call employee.

This finding aligns with Adebayo et al. (2025), who described AI's strongest construction applications as planning, monitoring, control, and decision-support tasks. In the interviews, AI was most valued when it reduced administrative burden, improved access to information, or supported repetitive processes. Participants were cautious about “flashy” tools that did not clearly solve a business problem.

Theme 2: Governance and Security Shape Adoption

Governance and data security were among the strongest themes, especially in the healthcare interview. The healthcare organization established an AI governance committee that includes senior leadership, providers, IT, and other organizational representatives. This committee reviews AI requests monthly, evaluates new AI features, and creates guardrails for responsible use. The healthcare participant emphasized that HIPAA compliance and protection of patient health information are central to every AI decision.

Security also shaped AI adoption at General Contractor B. The senior vice president explained that tool selection was heavily influenced by how company data would be handled and whether proprietary information could be used to train outside models. General Contractor A also selected Microsoft Copilot partly because it fit within the Microsoft 365 environment and was viewed as a safer option for confidentiality. These findings match Mohamed et al. (2025), who identified security, data, performance, and governance risks as major issues in generative AI adoption.

Theme 3: Implementation Is Gradual and Pilot-Based

All organizations described staged implementation rather than rapid, company-wide adoption. The healthcare organization described its approach as “crawl, walk, run,” beginning with governance review, then pilots, then KPIs, and finally broader scaling. General Contractor A created an AI focus group and an AI implementation team. The focus group identifies workflow problems, while the implementation team evaluates whether to build or buy a solution. The company is also rolling out Copilot on selected pilot projects with tech-forward teams before broader adoption.

General Contractor B similarly described a slow and steady approach that begins with low-risk use cases and expands after value is demonstrated. The project manager from General Contractor A described a similar pattern during the Procore rollout, where a small committee of users from multiple departments became internal experts before the platform was expanded company-wide. These findings align strongly with Omotayo et al. (2025), who emphasized staged maturity, training, readiness, and cultural attitudes as factors that shape AI-powered technology adoption.

Theme 4: AI Is Currently Strongest in Office-Based and Document-Heavy Workflows

The interviews suggest that AI is currently creating the clearest value in office-based workflows. Examples included summarizing email chains, generating meeting notes, reviewing contracts and subcontracts, searching specifications, locating drawing details, drafting provider responses, supporting clinical documentation, and answering internal process questions. These workflows are often repetitive, text-heavy, and information-intensive, which makes them well suited for current AI tools.

The project manager from General Contractor A explained that Copilot supports meeting minutes by capturing discussion points and action items in the background. Procore's AI search function can also reduce the time required to locate specification sections, product information, or drawing details. General Contractor B reported time savings on subcontract review and internal process searches. The healthcare organization reported measurable savings from internally developed IT agents. These examples support the TAM framework because participants valued AI when it was clearly useful and easy to integrate into existing work practices (Na et al., 2023).

Theme 5: Field Adoption Remains More Complex

Although participants saw strong potential for AI in field operations, they also described field adoption as more difficult than office adoption. General Contractor B noted that the field side has not benefited as much yet because construction field work is physical, data is messy, and the margin for error is smaller. General Contractor A described construction as a traditional industry that is slow to adopt new technologies, especially among experienced field personnel who may be skeptical of new tools.

However, the interviews also showed promising field-related use cases. General Contractor A is using reality capture tools such as 360-degree photos, drone footage, laser scans,

and AI-supported safety observations. The project manager described a safety example where a 360 camera identified a worker without proper PPE and connected the issue to an OSHA-related standard. The construction technology director also described future AI progress tracking, where 360 photos or laser scans could be compared with BIM models to estimate completion percentages for work such as ductwork or insulation. These examples suggest that field adoption may grow as AI becomes more embedded in existing reality capture workflows.

Theme 6: Training, Trust, and Human Oversight Remain Essential

Across interviews, participants emphasized that AI output cannot simply be accepted without human review. In the healthcare organization, Epic-generated draft responses to patient questions still require providers to read, edit, and approve messages before sending them. In construction, Copilot meeting minutes are used as a supplement rather than a replacement for human note-taking. Procore AI search and safety observations also require professional judgment to verify accuracy and context.

Training was another important issue. General Contractor A noted that teaching anyone a new process creates challenges, especially when time and money are limited. The project manager explained that monthly Procore training and office hours helped employees learn specific platform functions. General Contractor B focused on helping employees use Copilot correctly while allowing a smaller group to explore more advanced agent-based use cases. These findings support Na et al. (2023) because ease of use, trust, and compatibility with existing work are central to user acceptance.

Discussion

The findings support the literature's argument that AI adoption is not simply a technical issue. Instead, it is an organizational change process shaped by trust, governance, readiness,

leadership, and workflow fit. The interviews show that organizations are not adopting AI all at once. They are experimenting selectively, often through committees, pilots, and internal champions. This supports capability maturity thinking because AI adoption appears to move through stages: initial exploration, pilot testing, controlled implementation, measurement, and scaling.

The interviews also demonstrate the importance of perceived usefulness. Participants were most positive about AI when it saved time, reduced repetitive work, or improved access to information. Examples included meeting summaries, specification searches, subcontract review, help desk automation, provider documentation, and on-call lookup. These are not futuristic or abstract use cases; they are practical workflow improvements. This reinforces TAM by showing that employees are more likely to accept AI when it is useful, easy to access, and compatible with existing platforms such as Microsoft 365, Procore, Epic, and DroneDeploy.

A second important discussion point is the difference between office and field adoption. AI is currently more mature in administrative, document-heavy, and information-based work. Field use cases are emerging, particularly in safety monitoring and reality capture, but they face additional barriers. Field work involves physical conditions, incomplete or messy data, higher safety consequences, and stronger dependence on craft knowledge and superintendent judgment. As a result, field AI adoption may require more time, better data capture, and greater trust-building with field teams.

The healthcare interview also expands the construction-focused discussion by showing how owner/facilities organizations may adopt AI differently from contractors. In the healthcare setting, governance and compliance are central because AI tools may interact with patient data. However, the organization's use of internal agents for IT support also shows that facilities-

related organizations can benefit from AI in operational workflows outside direct patient care. This is relevant to construction and facilities management because owners, hospitals, and facility operators may increasingly expect contractors and consultants to understand AI governance, data handling, and integration with owner systems.

Overall, the findings suggest that AI adoption in construction and facilities is best understood as a cautious but accelerating shift. Organizations are not abandoning human expertise. Instead, they are looking for ways to reduce administrative burden, improve information access, support decision-making, and create more efficient workflows while maintaining human oversight.

Conclusion and Recommendations

This study examined AI adoption across a healthcare facilities organization and two general contractors. The findings show that AI is already being used for meeting documentation, email and document summarization, specification search, subcontract review, clinical documentation, internal process support, safety monitoring, and early agent-based automation. However, adoption remains uneven. Office-based workflows are seeing clearer benefits than field workflows, while security, trust, cost, training, and data quality continue to shape implementation decisions.

Four major insights emerged. First, AI adoption is most effective when it is problem-driven rather than technology-driven. Second, governance and security are essential, especially when sensitive data or proprietary information may be involved. Third, successful implementation usually occurs through pilots, committees, and staged rollout rather than immediate company-wide deployment. Fourth, human oversight remains necessary because AI output must be reviewed, verified, and interpreted by trained professionals.

Based on these findings, construction and facilities organizations should consider the following recommendations:

- Start with a governance structure that includes leadership, technology staff, end users, and risk or compliance stakeholders.
- Identify a small number of high-value use cases before selecting tools. Strong starting points include meeting notes, specification search, document review, internal knowledge support, and safety observation workflows.
- Use pilot projects and internal champions before broad rollout. Early users should help test workflows, identify problems, and train others.
- Measure value through practical KPIs such as time saved, ticket reductions, meeting efficiency, documentation quality, safety observations, or reduced rework.
- Maintain human oversight and clear review procedures, especially for clinical, safety, contractual, or field-related outputs.

AI will likely become more embedded in construction and facilities work over the next five years, especially through agents, reality capture, progress tracking, document review, and operational automation. Organizations that develop strong governance, reliable data practices, and user-centered implementation strategies will be better positioned to benefit from AI while managing its risks.

References

Adebayo, Y., Udoh, P., Kamudyariwa, X. B., & Osobajo, O. A. (2025). Artificial intelligence in construction project management: A structured literature review of its evolution in application and future trends. *Digital*, 5(3), 26. <https://doi.org/10.3390/digital5030026>

- Mohamed, M. A. H., Al-Mhdawi, M. K. S., Ojiako, U., Dacre, N., Qazi, A., & Rahimian, F. (2025). Generative AI in construction risk management: A bibliometric analysis of the associated benefits and risks. *Urbanization, Sustainability and Society*, 2(1), 198-230. <https://doi.org/10.1108/USS-11-2024-0069>
- Na, S., Heo, S., Choi, W., Kim, C., & Whang, S. W. (2023). Artificial intelligence (AI)-based technology adoption in the construction industry: A cross-national perspective using the Technology Acceptance Model. *Buildings*, 13(10), 2518. <https://doi.org/10.3390/buildings13102518>
- Omotayo, T., Tanyer, A. M., Deng, J., Kaima, A., Akponeware, A., Pekerikli, M. K., Shikder, S., & Ogunnusi, M. (2025). Advancing AI-powered BIM for circularity in construction in the UK and Turkiye: State-of-the-art review and capability maturity modelling. *Buildings*, 15, 1224. <https://doi.org/10.3390/buildings15081224>

Appendix A

Interview Protocol

The interviews were guided by the assignment questionnaire. Questions addressed participant role and responsibilities, how the organization is using or exploring AI, tools and platforms used, evaluation and selection, leadership involvement, training, barriers, outcomes, strategic vision, and advice for other firms beginning their AI journey.

Core Questions

- Please describe your role and responsibilities.
- How is your organization currently using or exploring AI?
- What AI tools or platforms are currently used?
- How were these tools evaluated and selected?
- How is leadership involved in AI adoption?
- What training or upskilling has been provided?
- What are the major obstacles your organization faced while adopting AI?
- Were there cultural, technical, or financial challenges?
- What benefits have you observed after implementing or testing AI?
- Are there measurable results, such as time, cost, or quality improvements?
- How do you see AI shaping your organization or the industry in the next five years?
- What advice would you offer to other firms beginning their AI journey?

Appendix B

Interview Summary: Healthcare Organization - IT Director

The participant served as an IT director for applications integration. The organization has established a formal AI governance committee that includes senior leadership, providers, IT, and representatives from multiple areas. This committee reviews AI requests, evaluates new AI options, and creates guardrails for responsible use. The participant emphasized that healthcare AI adoption must address HIPAA compliance, PHI protection, and data security before tools can be widely implemented.

The organization uses AI through Epic, its primary electronic medical record system. AI can help draft responses to patient questions based on chart information, but providers must review and approve the response before it is sent. The organization also uses ambient clinical documentation, where an application listens during patient-provider conversations and generates medical documentation. The participant stated that this improves efficiency by reducing provider documentation burden after appointments.

The organization has also developed internal AI agents using Microsoft Copilot Studio. Examples include an agent that can reboot an employee's assigned computer after validating the user and device, and another that can identify who is on call for a specific team and provide contact information. The participant estimated that the computer reboot function saves approximately nine hours per week and that the on-call lookup feature saves approximately nine hours per week. The participant described the AI strategy as a crawl, walk, run approach using governance review, pilots, KPIs, and gradual scaling.

Appendix C

Interview Summary: General Contractor A - Director of Construction Technology

The participant served as a director of construction technology for a general contractor. His role included technology beyond standard IT support, including drones, 360-degree photos, laser scans, BIM coordination, and photorealistic renderings. The company began more formal AI exploration after seeing larger firms and industry conferences focus increasingly on AI.

The participant emphasized that AI should not be treated as a solution by itself. Instead, AI should be connected to a real problem, workflow pain point, or tedious task. The company created an AI focus group to identify needs and an AI implementation team to evaluate solutions.

The participant described a build-versus-buy decision process when considering whether to develop internal tools or purchase commercial AI platforms.

The company uses Microsoft Copilot because it integrates with Microsoft 365 and provides a safer environment for confidentiality. The company is piloting Copilot on selected projects with tech-forward teams and evaluating results over time. Other AI-related uses include meeting transcription and action-item tools, DroneDeploy, 360-degree photos, drones, and safety AI that can identify potential PPE violations. Looking forward, the participant expects AI to support progress tracking by comparing reality capture data with BIM models to estimate completion status and improve subcontractor coordination.

Appendix D

Interview Summary: General Contractor A - Project Manager

The participant was a project manager responsible for office-side project management tasks such as submittals, pay applications, RFIs, document control, drawings, subcontractor coordination, scheduling, material release, and support for the field superintendent. From this role, AI was discussed primarily as a project management and documentation support tool.

The participant described using Microsoft Copilot as a supplement for meeting minutes during owner-architect-contractor meetings and subcontractor meetings. Copilot can help capture discussion points, action items, and commitments, but the participant emphasized that human review is still required because AI can misidentify speakers or miss context. The participant also described using Procore AI/search functions to locate specification sections, submittal information, product data, and drawing details more efficiently than manually searching through long PDF documents.

The participant explained that leadership was involved in the Procore rollout because of cost, training needs, and the scale of implementation. The company used training sessions and monthly office hours to help staff learn specific functions. The participant also described field-side resistance as a normal part of adopting new software, especially for superintendents and field users working from tablets or iPads. Future AI use was connected to safety monitoring through 360 cameras and automatic identification of potential PPE issues.

Appendix E

Interview Summary: General Contractor B - Senior Vice President / Technology Group

Leader

The participant was a senior vice president at a mid-sized general contractor with approximately 350 employees. The participant had a technology background and was involved in a technology group leading process improvement and AI-related initiatives. The organization uses AI for research, quick answers, subcontract review, and internal process support.

The participant stated that the company uses Microsoft Copilot company-wide and that a smaller group has approval to use Claude Code for software development and process improvement.

Security and data handling were the primary factors in tool evaluation. The company wanted confidence that proprietary information would not be exposed or used to train outside models.

Leadership support was described as strong, with AI adoption framed as a company initiative rather than a grassroots effort. Training has focused on helping employees use Copilot properly, while a smaller group explores more advanced agent-based workflows. The participant stated that AI is saving time on document-heavy work, especially subcontract review, and that internal process agents are reducing time spent searching for answers about company procedures.

However, the participant noted that field-side construction has not benefited as much yet because the work is physical, data is messy, and error margins are smaller.